

# Two-Sample Inference With Long-Term Survivors

A comparative simulation study and new two-sample test statistics

Durbadal Ghosh  
with Yu Bi and Subodh Selukar

March 23, 2026

# Modern Oncology Now Creates Long-Term Survivors

## Therapy effect

50%

5-year OS with combination checkpoint  
therapy

44%

5-year OS with monotherapy

## Population effect

42

cancer sites with documented long-term  
survivorship patterns

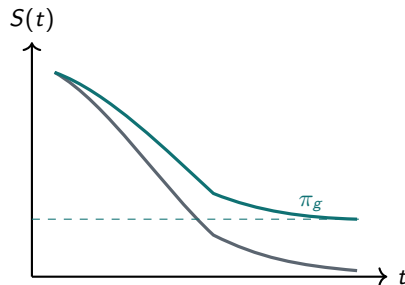
**Take-home:** Improved therapy changes the data-generating mechanism, not only the clinical outcome rate.

Numbers taken from the introduction of the comparative simulation study.

# What We Mean by Long-Term Survivors

$$S_g(t) = \pi_g + (1 - \pi_g)S_{u,g}(t)$$

- group  $g \in \{0, 1\}$
- $\pi_g$ : L-TS / cure fraction
- $S_{u,g}(t)$ : survival among uncured subjects
- $S_g(t) \rightarrow \pi_g$  as  $t \rightarrow \infty$



# Why Cure Matters Clinically

## Patient care

- surveillance intensity
- survivorship planning
- transition from oncology follow-up to primary care

## Health systems

- resource allocation
- imaging burden and anxiety
- long-term value assessment of immunotherapies

If we can identify who is effectively cured, both clinical follow-up and policy decisions change. Clinical motivation synthesized from v3.tex and the comparative study introduction.

# Conventional Survival Analysis Assumes No L-TS

## Implicit assumption

all subjects eventually fail

The ordinary log-rank framework is built for this regime.

## When L-TS exist

- the assumption is false
- $S_g(t)$  does not go to 0
- hazard-based summaries may no longer target the clinical effect of interest

Comparative study introduction: data with L-TS violate the fundamental assumption that all subjects eventually experience the event.

# Why Cure Modeling Exists

## Incidence component

$$\pi_g$$

- who is long-term event free?
- durable response
- cure probability

## Latency component

$$S_{u,g}(t)$$

- when do uncured subjects fail?
- short-term progression
- timing of events

**Take-home:** Cure modeling separates two scientific questions that a single hazard ratio cannot separate.

# Statistical Reason 1: Estimand Mismatch

| Observed difference     | What may be changing      | What a single hazard-based summary misses |
|-------------------------|---------------------------|-------------------------------------------|
| Late plateau separation | $\pi_1 - \pi_0$           | incidence effect                          |
| Early separation only   | $S_{u,1}(t) - S_{u,0}(t)$ | uncured-survival effect                   |
| Both                    | $\pi_g$ and $S_{u,g}$     | mechanism-specific interpretation         |

**Take-home:** If treatment acts through cure fraction, uncured survival, or both, the inferential target should reflect that structure.

## Statistical Reason 2: Hazard-Based Signal Can Vanish

$$h_g(t) = \frac{(1 - \pi_g)f_{u,g}(t)}{S_g(t)} \implies h_g(t) \rightarrow 0 \text{ as } t \rightarrow \infty$$

- Log-rank and weighted log-rank tests are functions of  $h_1(t) - h_0(t)$
- With L-TS in both groups, both hazards decay toward 0
- So late follow-up can reduce the hazard-based signal even when survival curves remain clinically different

This is the central explanation in the discussion section of the comparative study.

# Statistical Reason 3: Follow-Up Can Bias Cure Inference

## What the paper states

Insufficient follow-up can induce **unpredictable bias** in cure-model estimates.

## Why

- plateau not yet identified
- cure and long uncured survival are confounded
- model choice can be unstable

Misestimating  $\pi_g$  directly biases long-term survival prediction and can mislead prognosis.

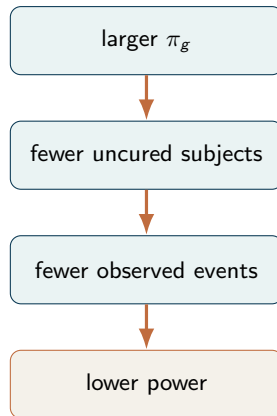
**Take-home:** Cure modeling is needed, but it is also maturity-sensitive.

Comparative study introduction; see also Othus et al. (2020), Bias in Mean Survival From Fitting Cure Models With Limited Follow-Up.

# Statistical Reason 4: Higher Cure Means Fewer Informative Events

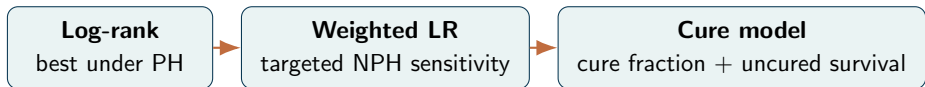
## Information argument

- larger  $\pi_g$  means fewer uncured subjects
- fewer uncured subjects means fewer events
- fewer events mean less information for two-sample testing



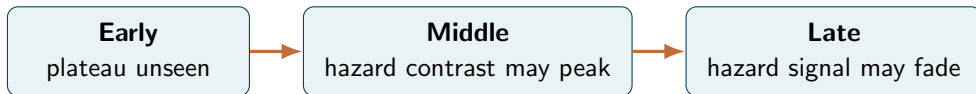
This explanation is made explicit in the supplement for high-cure scenarios.

# Challenge 1: Which Test Should a Trial Use?



- The right choice depends on whether the signal is hazard-based, cure-based, or mixed
- That is exactly the challenge the comparative study addresses

## Challenge 2: How Much Follow-Up Is Enough?



- More follow-up is not automatically more power for all tests
- That follow-up dependence is a core result of the comparative study

## Challenge 3: Is the Treatment Effect on Cure, Latency, or Both?

Cure only

$$\pi_1 \neq \pi_0, \quad S_{u,1} = S_{u,0}$$

Latency only

$$\pi_1 = \pi_0, \quad S_{u,1} \neq S_{u,0}$$

Both

$$\pi_1 \neq \pi_0, \quad S_{u,1} \neq S_{u,0}$$

**Take-home:** The method should be able to respond to whichever mechanism is present.

# What Existing Methods Target

| Method             | Primary target                | Strength                 | Weakness               |
|--------------------|-------------------------------|--------------------------|------------------------|
| Log-rank           | hazard difference             | optimal under PH         | ignores L-TS structure |
| Weighted LR        | time-local hazard difference  | flexible under NPH       | weight choice matters  |
| Adaptive YP / TS   | unknown NPH pattern           | adaptive hazard emphasis | still hazard-centered  |
| Mixture cure model | $\pi_g$ and $S_{u,g}$ jointly | mechanism-aware          | needs maturity / model |

# The Gap the Comparative Study Addresses

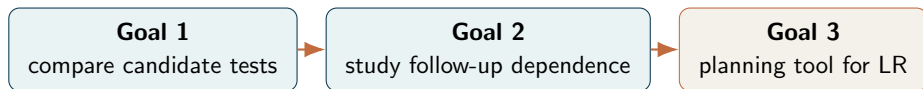
## What was already known

- NPH test comparisons
- cure-model estimation
- cure-model adequacy rules

## What was missing

- broad two-sample test comparison with L-TS
- explicit study of follow-up time
- practical prediction of non-monotone log-rank power

# Goals of the Comparative Simulation Study



# Comparative Study: Notation and Hypothesis

$$H_0 : S_0(t) = S_1(t) \text{ for all } t \geq 0 \quad \text{vs.} \quad H_a : S_0(t) \neq S_1(t) \text{ for some } t \geq 0$$

- control group:  $g = 0$
- treatment group:  $g = 1$
- cure fraction:  $\pi_g$
- uncured survival:  $S_{u,g}(t)$
- overall survival:  $S_g(t)$
- follow-up horizon:  $\tau$
- cure effect:  $OR$
- uncured-survival effect:  $HR_u$

# Candidate Methods: Hazard-Based

| Method      | Statistic / idea                          | Best-case regime        |
|-------------|-------------------------------------------|-------------------------|
| LR          | unweighted log-rank                       | PH, no L-TS             |
| Early WLR   | FH $G^{1,0}$                              | early hazard separation |
| Late WLR    | FH $G^{0,1}$                              | late hazard separation  |
| Optimal LR  | weight proportional to log-hazard ratio   | known hazard pattern    |
| Adaptive YP | short-term / long-term hazard ratio model | broad NPH               |
| Adaptive TS | log-rank + adaptive weighted stage        | selected NPH patterns   |

# Candidate Methods: Cure-Aware and Planning-Oriented

## Mixture cure model

$$S_g(t) = \pi_g + (1 - \pi_g)S_{u,g}(t)$$

- correctly specified parametric LRT benchmark
- directly models cure fraction + uncured survival

## Planning quantity

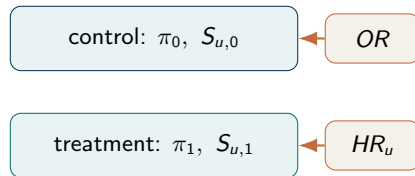
$$A(\tau) = \int_0^\tau w(t)\{h_1(t) - h_0(t)\} dt$$

- predicts shape of LR power over follow-up
- derived from expected LR score

# Data-Generating Mechanism

$$S_g(t) = \pi_g + (1 - \pi_g)S_{u,g}(t)$$

- control arm defined by  $\pi_0$  and  $S_{u,0}$
- treatment differences encoded separately on cure fraction and uncured survival
- all simulation scenarios are built from this mixture cure structure



# How Treatment Effects Were Encoded

$$\begin{aligned} \text{logit}(\pi_1) - \text{logit}(\pi_0) &= \log(OR) \\ h_{u1}(t) = HR_u h_{u0}(t) &\implies S_{u,1}(t) = \{S_{u,0}(t)\}^{HR_u} \end{aligned}$$

- $OR > 1$ : treatment improves the cure fraction
- $HR_u < 1$ : treatment improves uncured survival
- the paper studies settings where treatment survival is uniformly no worse than control

# Observed Data, Accrual, and Censoring

- 1 Simulate true event time  $T$  from the mixture cure model
- 2 Generate censoring under calendar-time accrual
- 3 Observe

$$Y = \min(T, C), \quad \delta = I(T \leq C)$$

## Design detail

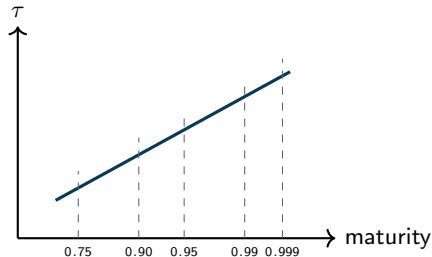
Administrative censoring uses a fixed accrual duration equal to half of the 75th percentile of the control-group uncured distribution.

# Study Design Grid

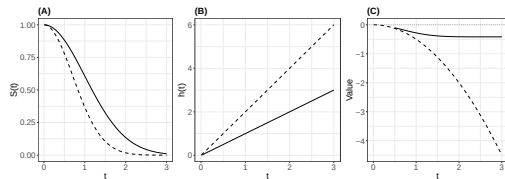
| Component             | Values                                                                                    |
|-----------------------|-------------------------------------------------------------------------------------------|
| Uncured families      | Weibull, gamma, log-logistic                                                              |
| Control cure fraction | $\pi_0 \in \{0, 0.1, 0.2, 0.5, 0.8, 0.95\}$                                               |
| Cure effect           | $OR \in \{1.0, 1.1, 1.25, 1.5\}$                                                          |
| Latency effect        | $HR_u \in \{1.0, 0.9, 0.75, 0.5\}$ for Weibull; corresponding TR for gamma / log-logistic |
| Sample size           | $n = 25, 50, 100, 500$ per arm                                                            |
| Follow-up             | control-uncured quantiles 0.75, 0.90, 0.95, 0.99, 0.999                                   |
| Replicates            | 10,000 datasets per setting                                                               |

# Why Follow-Up Was Indexed by Quantiles

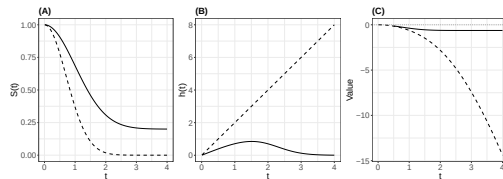
- quantiles map maturity across very different survival families
- a 95th percentile analysis means “very mature” across settings
- this makes pattern comparison easier than raw calendar time



# Scenario Gallery I

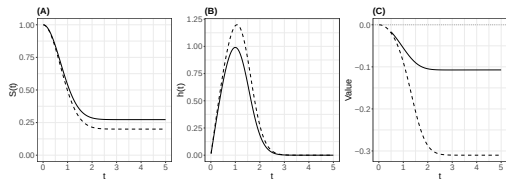


left: no L-TS in either arm

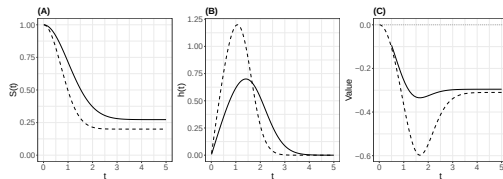


right: L-TS in treatment arm only

# Scenario Gallery II



left: cure effect only



right: cure + uncured-survival effects

# How Performance Was Measured

$$\widehat{\text{reject}} = \frac{1}{R} \sum_{r=1}^R I(p_r < 0.05)$$

Type I error

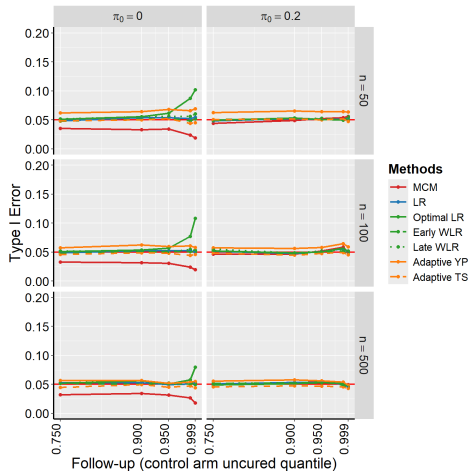
$$OR = 1, HR_u = 1$$

- compare across sample size
- compare across follow-up
- compare across effect mechanism

Power

$$OR > 1 \text{ or } HR_u < 1$$

# Type I Error Was Usually Controlled



# Type I Error: What Matters

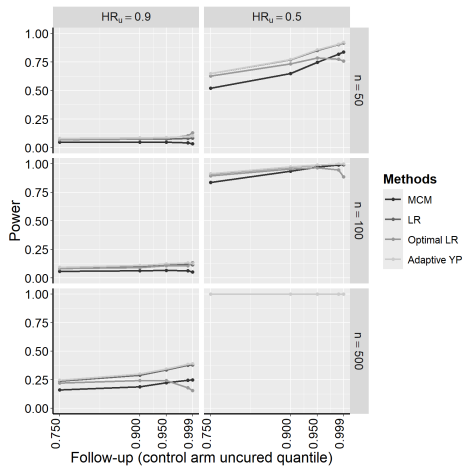
## Mostly good

- most methods stayed near 0.05
- control was stable across many cure settings

## Exceptions

- optimal LR inflated in no-cure / long follow-up settings
- adaptive YP showed modest inflation in some settings

# Results: No L-TS in Either Arm

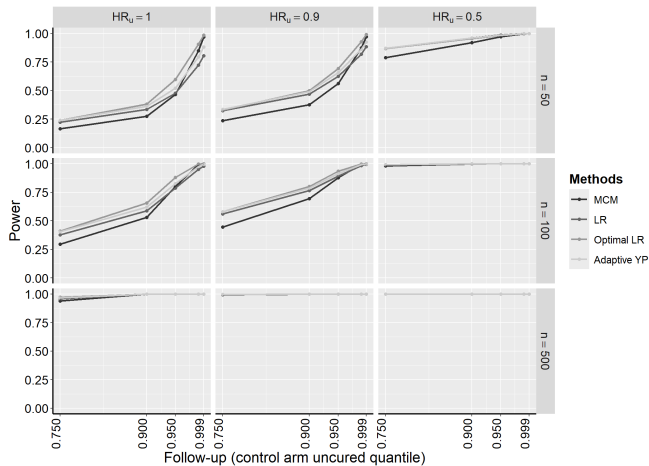


# No L-TS: Interpretation

- LR and adaptive YP usually had the highest power
- power increased with follow-up and sample size in the expected way
- the cure model did not help much when the data-generating mechanism had no cure

**Take-home:** When L-TS are absent, ordinary hazard-based tests remain difficult to beat.

# Results: L-TS in Treatment Arm Only

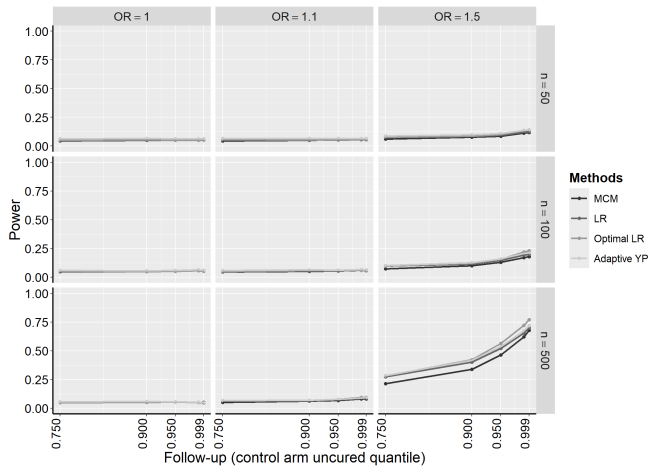


# Treatment-Only L-TS: Interpretation

- all methods improved as follow-up lengthened
- the cure model started weaker at shorter follow-up
- with enough maturity, cure-model performance approached the strongest hazard-based competitors

**Take-home:** The cure model needs maturity before its structural advantage becomes visible.

# Results: L-TS in Both Arms, Cure Effect Only

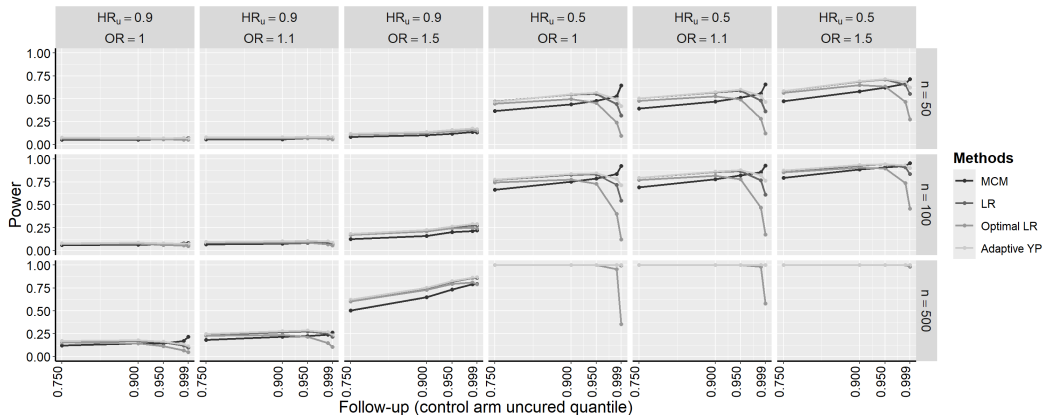


## Both Arms Have L-TS, Cure Effect Only

- even large *OR* values produced modest power
- pure cure-fraction differences are hard to detect
- hazard-based methods do not receive a large sustained signal in this regime

**Take-home:** This is the regime where “there is a cure effect” does not translate into easy testing.

# Results: L-TS in Both Arms, Cure and Latency Effects



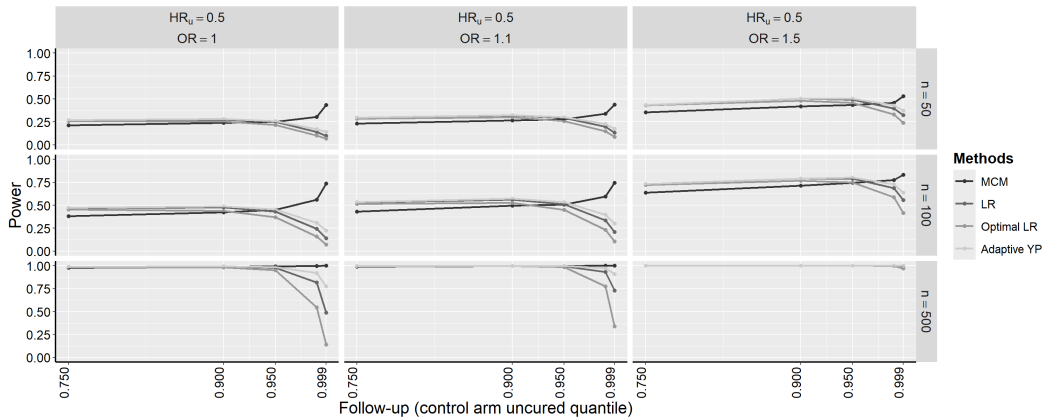
# The Key Comparative Result

| Method family               | Shape over follow-up | Best at longest follow-up? |
|-----------------------------|----------------------|----------------------------|
| LR / WLR / YP               | often non-monotone   | often no                   |
| Correctly specified MCM LRT | monotone increasing  | yes                        |

- hazard-based power can increase, peak, and then fall
- cure-model power can continue to accumulate with maturity

**Take-home:** More follow-up can help one method and hurt another on the same scientific question.

# Higher Cure Fractions Reduce Power Across the Board



# Why Hazard-Based Methods Behave So Similarly

$$U(\tau) = \int_0^\tau \left\{ dn_1(t) - \frac{Y_1(t)}{Y(t)} dn(t) \right\}$$

all hazard-based candidates are functions of  $h_1(t) - h_0(t)$

- different weights change emphasis
- but they all depend on the same vanishing late hazard signal
- this explains the shared non-monotone pattern when both groups have L-TS

# Expected Log-Rank Signal

$$\mathbb{E}\{U(\tau)\} \approx \int_0^\tau w(t)\{h_1(t) - h_0(t)\} dt$$
$$w(t) = G(t) \frac{2S_0(t)S_1(t)}{S_0(t) + S_1(t)}$$

- large-sample approximation
- balanced allocation
- censoring enters through  $G(t)$

# Weighted Average Hazard Difference

$$A(\tau) = \int_0^{\tau} w(t) \{h_1(t) - h_0(t)\} dt \quad \Delta H(t) = H_1(t) - H_0(t)$$

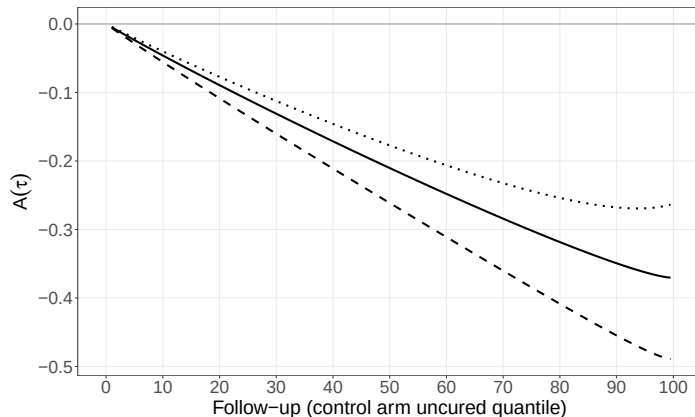
If  $A(\tau)$  moves away from 0

- LR signal strengthens
- power tends to rise

If  $A(\tau)$  turns back

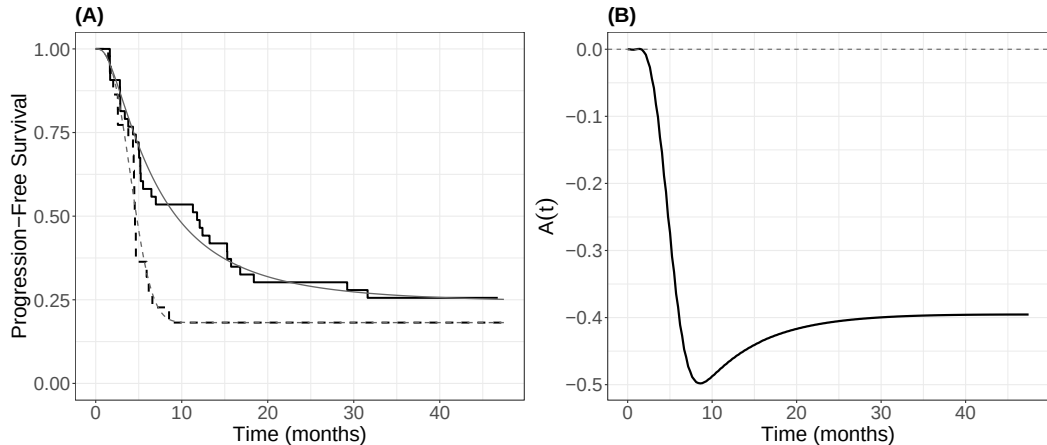
- LR signal weakens
- power can flatten or fall

# Prediction and Empirical Power Match Well



**Take-home:** Monotone  $A(\tau)$  predicts monotone power; non-monotone  $A(\tau)$  predicts non-monotone power.

# Data Example: BEACON-Immuno Trial



The comparative study uses reconstructed BEACON-Immuno data to contextualize the simulation findings.

# BEACON Summary

## Observed $p$ -values

| Method     | $p$   |
|------------|-------|
| LR         | 0.092 |
| FH(1,0)    | 0.051 |
| FH(0,1)    | 0.474 |
| Optimal LR | 0.116 |
| MCM LRT    | 0.079 |

## Cure-model fit

| Arm   | best family | $\hat{\pi}$ |
|-------|-------------|-------------|
| dB    | log-normal  | 0.24        |
| No dB | Weibull     | 0.18        |

$A(t)$  reached its minimum near 9 months, suggesting limited gain from much longer fixed-sample follow-up for LR.

# What the Comparative Study Implies for Trial Design

## Design

- choose test jointly with follow-up
- more time is not always more power
- cure fraction changes the information budget

## Analysis

- plateau does not guarantee one best test
- early analysis favors hazard methods more often
- late analysis can favor cure modeling

# Comparative Study: Summary

- L-TS change both the estimand and the power curve
- hazard-based tests can lose power at longer follow-up when both groups have L-TS
- correctly specified cure modeling benefits from maturity
- $A(\tau)$  provides a practical planning diagnostic for LR

# Why a Second Paper on New Test Statistics?

## From paper 1

- power depends on cure structure
- power depends on follow-up maturity
- forcing one test can be inefficient

## Paper 2 objective

- build adaptive two-sample statistics
- use cure models only when data support them
- calibrate the full rule, not only a branch

**Take-home:** The second paper targets the testing problem that the comparative study exposes.

# Observed Data and Mixture-Cure Notation

- subject  $i$ :  $(Y_i, \delta_i, a_i)$
- arm indicator  $a_i \in \{0, 1\}$
- $Y_i = \min(T_i, C_i)$
- $\delta_i = I(T_i \leq C_i)$

$$S(t \mid a_i) = \pi_i + (1 - \pi_i)S_u(t \mid a_i)$$

- $\pi_g$ : cure fraction in arm  $g$
- $S_{u,g}(\cdot)$ : uncured survival in arm  $g$
- cure effect is summarized by  $OR$
- uncured effect is summarized by  $HR_u$

$$\text{logit}(\pi_1) - \text{logit}(\pi_0) = \log(OR)$$

$$S_{u,1}(t) = [S_{u,0}(t)]^{HR_u}$$

Inside each working model, family-specific arm coefficients encode these two scientific effects.

# Uncured Survival Families Used in Both Approaches

| Family      | Uncured survival in arm $g$                                       | Arm-specific uncured-distribution parameters                         |
|-------------|-------------------------------------------------------------------|----------------------------------------------------------------------|
| Exponential | $S_{u,g}(t) = e^{-r_g t}$                                         | $r_g = e^{\alpha_r + \beta_r g}$                                     |
| Weibull     | $S_{u,g}(t) = \exp\left\{-(t/\lambda_g)^k\right\}$                | $k = e^{\alpha_k}, \lambda_g = e^{\alpha_\lambda + \beta_\lambda g}$ |
| Gamma       | $S_{u,g}(t) = 1 - F_\Gamma(t; k, r_g)$                            | $k = e^{\alpha_k}, r_g = e^{\alpha_r + \beta_r g}$                   |
| Lognormal   | $S_{u,g}(t) = 1 - \Phi\left(\frac{\log t - \mu_g}{\sigma}\right)$ | $\mu_g = \mu + \beta_\mu g, \sigma = e^{\alpha_\sigma}$              |
| Gen. gamma  | $S_{u,g}(t) = S_{GG}(t; \mu_g, \sigma, Q)$                        | $\mu_g = \mu + \beta_\mu g, \sigma = e^{\alpha_\sigma}$              |

Across both methods, the cure effect is indexed by  $OR$  and the uncured treatment effect by  $HR_u$ .

The family-specific coefficients  $\beta_r, \beta_\lambda, \beta_\mu$  are working-model parameters for that uncured effect.

# Running Example: Generalized-Gamma Cure Model

$$\eta^{(GG)} = (\theta, \mu, \alpha_\sigma, Q, \beta_\pi, \beta_\mu)^\top$$

$$\pi_i = \text{expit}(\theta + \beta_\pi a_i)$$

$$\mu_i = \mu + \beta_\mu a_i, \quad \sigma = e^{\alpha_\sigma}$$

$$T_{u,i} \mid a_i \sim GG(\mu_i, \sigma, Q)$$

$$H_0 : (\beta_\pi, \beta_\mu) = (0, 0)$$

$$H_1 : (\beta_\pi, \beta_\mu) \neq (0, 0)$$

$$S_{GG,i}(t) = S_{GG}(t; \mu_i, \sigma, Q)$$

$$f_{GG,i}(t) = f_{GG}(t; \mu_i, \sigma, Q)$$

- scientific treatment effects are still *OR* for cure and  $HR_u$  for uncured survival
- same architecture extends to exponential, Weibull, gamma, and lognormal
- in the talk, generalized gamma is the clearest running example for the cure branch

# Likelihood Used by the Cure Branch

$$\ell^{(d)}(\eta) = \sum_{i=1}^n \ell_i^{(d)}(\eta)$$

$$\ell_i^{(d)}(\eta) = \delta_i \log\{(1 - \pi_i)f_{u,i}^{(d)}(Y_i)\} + (1 - \delta_i) \log\{\pi_i + (1 - \pi_i)S_{u,i}^{(d)}(Y_i)\}$$

## Event contribution

$$L_i = (1 - \pi_i)f_{u,i}^{(d)}(Y_i), \quad \delta_i = 1$$

## Censoring contribution

$$L_i = \pi_i + (1 - \pi_i)S_{u,i}^{(d)}(Y_i), \quad \delta_i = 0$$

**Take-home:** Only the uncured-survival family  $d$  changes; the cure link and tested arm-effect structure stay the same.

# RECeUS Screening: Arm-Level Rule

## Candidate family set

$$\mathcal{D} = \{\text{Exp, Weibull, Gamma, LN, GG}\}$$

$$\hat{d}_a = \arg \min_{d \in \mathcal{D}} AIC_{a,d}$$

## For arm $a \in \{0, 1\}$

- require at least 3 events
- fit 5 one-arm cure families
- choose  $\hat{d}_a$  by smallest AIC
- compute  $\hat{\pi}_a$  and  $\hat{S}_a(\tau_R)$

$$\hat{S}_{u,a}(\tau_R) = \frac{\hat{S}_a(\tau_R) - \hat{\pi}_a}{1 - \hat{\pi}_a}, \quad \hat{r}_{n,a} = \frac{\hat{S}_{u,a}(\tau_R)}{\hat{S}_a(\tau_R)}$$

$$\tau_R = \max_i Y_i$$

$$G_{n,a} = I(\hat{\pi}_a > 0.025, \hat{r}_{n,a} < 0.05)$$

Arm  $a$  passes if  $G_{n,a} = 1$

$$\hat{r}_{n,a} = \frac{\hat{S}_{u,a}(\tau_R)}{\hat{S}_a(\tau_R)}$$

equivalently  $\hat{\pi}_a > 0.025, \quad \hat{r}_{n,a} < 0.05$

# RECeUS Screening: Dataset-Level Selection

## Dataset-level gate

$$G_n = I(G_{n,0} + G_{n,1} \geq 1)$$

$G_n = 1 \implies$  cure branch allowed

$G_n = 0 \implies$  use non-cure branch

## Selected family on the cure branch

$D_n$  is defined only when  $G_n = 1$

$$D_n = \begin{cases} \hat{d}_0, & G_{n,0} = 1, G_{n,1} = 0, \\ \hat{d}_1, & G_{n,0} = 0, G_{n,1} = 1, \\ \hat{d}_{a^*}, & G_{n,0} = G_{n,1} = 1, \end{cases}$$

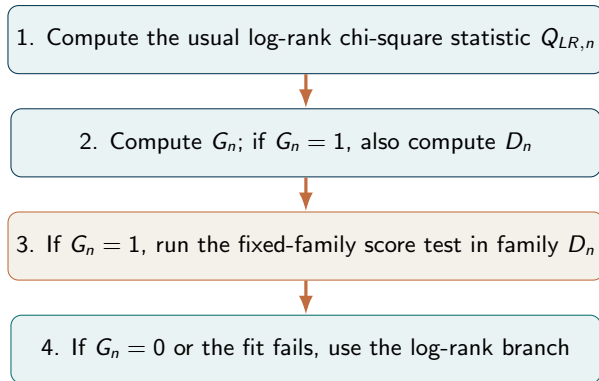
$$a^* = \arg \min_{a \in \{0,1\}} AIC_{a, \hat{d}_a}$$

$$\mathcal{S}_n = \{G_n = 1 \text{ and the selected-family score fit exists}\}$$

$$\mathcal{A}_{n,d} = \mathcal{S}_n \cap \{D_n = d\}, \quad \mathcal{B}_n = \mathcal{S}_n^c$$

**Take-home:**  $G_n$  opens the cure branch,  $D_n$  labels the chosen family on that branch,  $\mathcal{S}_n$  means the branch can actually be fitted, and  $\mathcal{B}_n$  is the fallback event.

## Method 1: Adaptive Conditional Score Test



$$Q_{1,n} = I(\mathcal{B}_n) Q_{LR,n} + \sum_{d \in \mathcal{D}} I(\mathcal{A}_{n,d}) Q_{\text{score},d,n}$$

# Method 1: Score Statistic Inside a Fixed Family

$$\eta^{(d)} = (\lambda^{(d)\top}, \psi^{(d)\top})^\top, \quad H_0 : \psi^{(d)} = 0$$

$\lambda^{(d)}$  = parameters estimated under  $H_0$ ,  $\psi^{(d)}$  = treatment-effect coefficients in family  $d$

$$\hat{\lambda}_{0,d} = \arg \max_{\lambda^{(d)}} \ell^{(d)}(\lambda^{(d)}, 0), \quad \tilde{\eta}_{0,d} = \left( \hat{\lambda}_{0,d}^\top, 0^\top \right)^\top$$

$$U_d(\eta) = \nabla \ell^{(d)}(\eta), \quad U_{r,d} = \nabla_{\psi} \ell^{(d)}(\tilde{\eta}_{0,d})$$

$$H_d(\tilde{\eta}_{0,d}) = \begin{pmatrix} H_{\lambda\lambda,d} & H_{\lambda\psi,d} \\ H_{\psi\lambda,d} & H_{\psi\psi,d} \end{pmatrix}, \quad H_d(\eta) = -\nabla^2 \ell^{(d)}(\eta)$$

$$I_{\psi|\lambda,d}(\tilde{\eta}_{0,d}) = H_{\psi\psi,d} - H_{\psi\lambda,d} H_{\lambda\lambda,d}^{-1} H_{\lambda\psi,d}$$

$$Q_{\text{score},d,n} = U_{r,d}^\top I_{\psi|\lambda,d}(\tilde{\eta}_{0,d})^{-1} U_{r,d}$$

**Take-home:** Every symbol is now explicit: score  $U_d$ , observed information  $H_d$ , and efficient information  $I_{\psi|\lambda,d}$ .

# Method 1: Generalized-Gamma Branch in Concrete Form

$$\eta^{(GG)} = (\theta, \mu, \alpha_\sigma, Q, \beta_\pi, \beta_\mu)^\top, \quad \psi^{(GG)} = (\beta_\pi, \beta_\mu)^\top$$

$$M_i = \pi_i + (1 - \pi_i)S_{GG,i}(Y_i)$$

$$u_{i,\beta_\pi} = a_i \begin{cases} -\pi_i, & \delta_i = 1, \\ \frac{\pi_i(1 - \pi_i)\{1 - S_{GG,i}(Y_i)\}}{M_i}, & \delta_i = 0 \end{cases}$$

$$u_{i,\beta_\mu}^{GG} = a_i \begin{cases} \frac{\partial \log f_{GG}(Y_i; \mu_i, \sigma, Q)}{\partial \mu_i}, & \delta_i = 1, \\ \frac{(1 - \pi_i) \partial S_{GG}(Y_i; \mu_i, \sigma, Q) / \partial \mu_i}{M_i}, & \delta_i = 0 \end{cases}$$

$$U_{r,GG} = \sum_{i=1}^n \begin{pmatrix} u_{i,\beta_\pi} \\ u_{i,\beta_\mu}^{GG} \end{pmatrix}$$

**Take-home:** For generalized gamma, the code gets the gradient and Hessian numerically from the full log-likelihood.

# Method 1: Conditional and Branch-Stable Null Laws

## Conditional cure-branch law

For any fixed  $d \in \mathcal{D}$  with  $\Pr(\mathcal{A}_{n,d}) > 0$  for large  $n$ ,

$$Q_{\text{score},d,n} \mid \mathcal{A}_{n,d} \Rightarrow \chi_2^2.$$

## Full adaptive statistic under branch-selection consistency

either  $\Pr(\mathcal{B}_n) \rightarrow 1$  or  $\exists d^* \in \mathcal{D}$  with  $\Pr(\mathcal{A}_{n,d^*}) \rightarrow 1$

$$\Pr(\mathcal{B}_n) \rightarrow 1 \implies Q_{1,n} \Rightarrow \chi_1^2$$

$$\Pr(\mathcal{A}_{n,d^*}) \rightarrow 1 \implies Q_{1,n} \Rightarrow \chi_2^2$$

**Take-home:** Here  $d^*$  is the unique family chosen with probability tending to 1. Boundary thresholds or AIC ties are the non-regular cases that can yield a mixture limit.

# Method 1: Proof Sketch for the Conditional Score Law

## Step 1: null estimator and score expansion

$$O_i = (Y_i, \delta_i, a_i)$$

$$0 = U_{\lambda,d}(\hat{\lambda}_{0,d}, 0)$$

$$\hat{\lambda}_{0,d} - \lambda_{0,d} = O_p(n^{-1/2})$$

$$n^{-1/2} U_{r,d}(\tilde{\eta}_{0,d}) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \phi_d(O_i) + o_p(1)$$

$$n^{-1} H_d(\tilde{\eta}_{0,d}) \rightarrow_p \mathcal{I}_d$$

$\phi_d(O_i)$  is the efficient-score influence function, and  $\mathcal{I}_d$  is the population efficient-information matrix.

## Step 2: why the branch stabilizes

- RECeUS thresholds are separated from the population values
- each arm has a unique smallest population AIC
- score-fit failure probability vanishes
- hence one branch event has probability tending to 1

## Step 3: branchwise reduction

$$Q_{1,n} - Q_{\text{score},d^*,n} = o_p(1),$$

on  $\mathcal{A}_{n,d^*}$

$$Q_{1,n} - Q_{LR,n} = o_p(1),$$

on  $\mathcal{B}_n$

**Take-home:** Method 1 adds one extra step beyond score theory: consistent RECeUS/AIC branch selection.

## Method 2: Standardized Maximum Test

### Fixed nonparametric components

- log-rank
- FH(1,0), FH(0,1), optimal
- RMST
- KM differences at fixed  $t_1, \dots, t_K$

### Fixed cure-aware component

- generalized-gamma score test
- null on  $(\beta_\pi, \beta_\mu)$
- no RECeUS and no family selection

$$\mathcal{M} = \{\text{LR}, \text{FH}(1,0), \text{FH}(0,1), \text{Opt}, \text{RMST}, \text{KM}(t_1), \dots, \text{KM}(t_K), \text{GG}\}$$

**Take-home:** Method 2 uses a fixed finite test vector, so the max-limit proof reduces to joint normality plus a continuous mapping.

## Method 2: Algorithm

1. Compute each raw component statistic in the fixed set  $\mathcal{M}$



2. Studentize to  $T_{m,n}$  using a consistent null standard error



3. If a component fails numerically, set  $|T_{m,n}| = 0$



4. Form  $Q_{2,n} = \max_{m \in \mathcal{M}} |T_{m,n}|$

## Method 2: Joint Asymptotic Normality

$$O_i = (Y_i, \delta_i, \mathbf{a}_i), \quad W_n = (T_{LR,n}, T_{FH10,n}, T_{FH01,n}, T_{opt,n}, T_{RMST,n}, T_{KM}(t_1), \dots, T_{KM}(t_K), T_{GG,n})^\top$$

with  $K < \infty$  fixed in advance.

- log-rank / FH / optimal: martingale score representations
- RMST / fixed-time KM: delta method from the KM process
- GG score: restricted-score linearization at  $\tilde{\eta}_{0,GG}$
- $\phi(O_i)$ : stacked influence function

$$W_n = \frac{1}{\sqrt{n}} \sum_{i=1}^n \phi(O_i) + o_p(1) \Rightarrow N(0, R), \quad R = \text{Var}\{\phi(O_1)\}$$

## Method 2: Max-Limit Proof Sketch

### Generalized-gamma score entry

$$T_{GG,n} = \left\{ n^{-1} l_{\psi|\lambda, GG}(\tilde{\eta}_{0, GG}) \right\}^{-1/2} \frac{1}{\sqrt{n}} U_{r, GG}(\tilde{\eta}_{0, GG}) + o_p(1)$$

$$\frac{1}{\sqrt{n}} U_{r, GG}(\tilde{\eta}_{0, GG}) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \phi_{GG}(O_i) + o_p(1), \quad T_{GG,n} \Rightarrow N(0, 1)$$

The same first-order structure holds for every other fixed component in  $\mathcal{M}$

### Maximum limit

$$Q_{2,n} = \max_{m \in \mathcal{M}} |T_{m,n}|$$

$$W_n \Rightarrow G \sim N(0, R) \quad \implies \quad Q_{2,n} \Rightarrow \max_{m \in \mathcal{M}} |G_m|$$

by Cramér-Wold, consistent studentization, and  $x \mapsto \max_m |x_m|$ .

# Assumptions Behind the Asymptotic Proofs

## Method 1: assumptions and role

- independent subjects + independent censoring: score CLT in each fixed family
- enough late follow-up: cure fraction and uncured-survival distribution are separately identifiable
- correct fixed-family cure model under  $H_0$ : score is centered at 0
- smooth interior parameter + nonsingular information: root- $n$  null estimator and  $\chi^2_2$  score limit
- separated RECeUS thresholds + unique population AIC winner:  $\Pr(\mathcal{B}_n) \rightarrow 1$  or  $\Pr(\mathcal{A}_{n,d^*}) \rightarrow 1$

## Method 2: assumptions and role

- independent subjects + independent censoring: joint CLT for LR/FH/RMST/KM pieces
- fixed finite component set  $\mathcal{M}$  and fixed  $t_1, \dots, t_K$ : finite-dimensional Gaussian limit
- correct GG null model + smooth interior parameter: linearization of the GG score
- finite second moments + consistent variance estimators: valid studentization
- continuous mapping for  $x \mapsto \max_m |x_m|$ : max-limit for  $Q_{2,n}$

**Take-home:** Method 1 and Method 2 share the basic CLT conditions, but only Method 1 needs consistency of the data-driven branch decision.

# Simulations Planned for the Two-Statistics Paper

## Core factors

- $\pi_0$ ,  $OR$ ,  $HR_u$ ,  $\tau$ ,  $n$
- one-arm and two-arm L-TS settings
- cure-family misspecification
- fixed true family vs selected family
- high-cure / sparse-event regimes

## Outputs

- type I error and power
- branch frequency for Statistic 1
- selected-family frequency for Statistic 1
- winner frequency for Statistic 2
- sensitivity to RECeUS, follow-up, and misspecification

# Data Application Placeholder 1

## Dataset summary

- disease / endpoint
- arm sizes
- KM curve with plateau
- follow-up distribution

## Model summary

- RECeUS result
- selected family  $D_n$
- fitted  $\hat{\pi}_g$  and uncured-survival parameters

**Take-home:** Reserved for the first real application once the analysis is complete.

# Data Application Placeholder 2

## Testing summary

- LR, MCM, Statistic 1, Statistic 2
- branch used by Statistic 1
- winner under Statistic 2

## Interpretation

- does signal look cure-driven?
- does follow-up change the conclusion?
- design implication for next study

**Take-home:** Reserved for the second application slide once the paper has a finalized dataset.

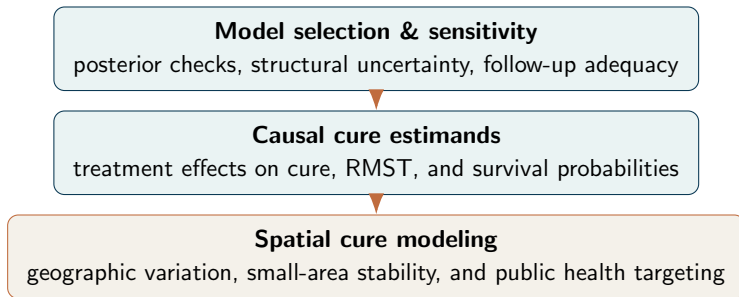
# Limitations of the Comparative Simulation Study

- no covariate adjustment or unequal allocation
- no crossing-survival scenarios
- focus on mixture cure formulations and selected parametric families
- follow-up indexed by control-uncured quantiles rather than study-specific calendar time
- very high-cure regimes remain difficult for all candidate tests

# Limitations of the Two-Sample Statistics Paper

- asymptotic arguments still need full formal write-up
- RECeUS gating can misclassify borderline datasets
- finite-sample behavior under strong misspecification needs extensive simulation
- practical performance will depend on how often the gate and family selection are correct
- application section is not yet populated with finalized real data

# Future Directions



Future-direction framing aligned to v3.tex.

## Scientific

- align inference with cure fraction + uncured-survival structure
- quantify when follow-up helps or hurts
- connect cure methodology with two-sample testing

## Practical

- better test choice in plateaued survival settings
- better sample-size / follow-up tradeoffs
- more defensible trial analysis plans

Questions

# Appendix: Assumptions Are Separate Conditions

| Label | Mathematical object constrained                                 | Why it is not implied by the others                                                            |
|-------|-----------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| A1    | joint law of $O_1, \dots, O_n$ : independence / iid structure   | does not determine censoring, model form, or information nonsingularity                        |
| A2    | relationship between $C_i$ and $T_i$ : $C_i \perp T_i \mid a_i$ | can hold or fail under the same sampling law and model form                                    |
| A3    | true survival model $S_g(t)$                                    | model correctness is separate from independence and follow-up adequacy                         |
| A4    | true parameter point and information matrix                     | interiority / $I(\eta_0) \succ 0$ are local regularity conditions, not sampling assumptions    |
| A5    | tail follow-up / identifiability condition                      | enough late follow-up is about observation horizon, not model correctness                      |
| A6    | RECeUS / AIC separation                                         | branch-selection consistency is about population thresholds and KL gaps, not about A1–A5 alone |

So the appendix uses A1–A6 as parallel conditions, not as a nested chain of implications.

# Appendix: Method 1 Assumptions in Mathematical Form

## Sampling, censoring, and fixed-family score regularity

$$O_i = (Y_i, \delta_i, a_i), \quad O_1, \dots, O_n \text{ independent}, \quad C_i \perp T_i \mid a_i$$

$$H_0 : \psi^{(d)} = 0, \quad \eta_0^{(d)} = (\lambda_0^{(d)\top}, 0^\top)^\top \in \text{int}(\Theta_d)$$

$$\mathbb{E}_{\eta_0^{(d)}} \left\| U_d(\eta_0^{(d)}) \right\|^2 < \infty, \quad n^{-1} H_d(\tilde{\eta}_{0,d}) \rightarrow_p \mathcal{I}_d, \quad \mathcal{I}_d \succ 0$$

## Branch-selection consistency

$$\exists d^* \in \mathcal{D} \text{ such that } \Pr(\mathcal{A}_{n,d^*}) \rightarrow 1 \quad \text{or} \quad \Pr(\mathcal{B}_n) \rightarrow 1$$

$$\Pr(\text{AIC ties}) \rightarrow 0, \quad \Pr(\text{selected-family score fit fails}) \rightarrow 0$$

## Why these assumptions matter

$$n^{-1/2} U_{r,d}(\tilde{\eta}_{0,d}) \Rightarrow N(0, \mathcal{I}_{\psi|\lambda,d}), \quad Q_{\text{score},d,n} \Rightarrow \chi_2^2$$

The first block gives the fixed-family score limit; the second block transfers that limit to  $Q_{1,n}$  by showing the adaptive branch decision becomes asymptotically deterministic.

# Appendix: Method 2 Assumptions in Mathematical Form

## Finite-dimensional joint CLT setup

$$\mathcal{M} = \{\text{LR}, \text{FH}(1,0), \text{FH}(0,1), \text{Opt}, \text{RMST}, \text{KM}(t_1), \dots, \text{KM}(t_K), \text{GG}\}, \quad K < \infty$$

$$T_{m,n} = \frac{1}{\sqrt{n}} \sum_{i=1}^n \phi_m(O_i) + o_p(1), \quad \mathbb{E}\{\phi_m(O_i)\} = 0, \quad \mathbb{E}\|\phi(O_i)\|^2 < \infty$$

$$R = \text{Var}\{\phi(O_1)\}, \quad \hat{R} \rightarrow_p R$$

## Generalized-gamma component

$$H_0 : (\beta_\pi, \beta_\mu) = (0, 0), \quad \eta_0^{(\text{GG})} \in \text{int}(\Theta_{\text{GG}}), \quad n^{-1} I_{\psi|\lambda, \text{GG}}(\tilde{\eta}_{0, \text{GG}}) \rightarrow_p \mathcal{I}_{\psi|\lambda, \text{GG}} \succ 0$$

## Why these assumptions matter

$$W_n = (T_{m,n} : m \in \mathcal{M})^\top \Rightarrow N(0, R)$$

$$Q_{2,n} = \max_{m \in \mathcal{M}} |T_{m,n}| \Rightarrow \max_{m \in \mathcal{M}} |G_m|, \quad G \sim N(0, R)$$

The finite- $K$  joint CLT gives the Gaussian vector, and the continuous mapping  $x \mapsto \max_m |x_m|$  gives the max-limit.

# Appendix: When the True Model Is Gamma Mixture Cure

## Simulation data-generating model

$$a_i \sim \text{Bernoulli}(1/2), \quad B_i \mid a_i = g \sim \text{Bernoulli}(1 - \pi_g)$$

$$T_i = \begin{cases} \infty, & B_i = 0, \\ T_{u,i}, & B_i = 1, \end{cases} \quad T_{u,i} \mid a_i = g \sim \Gamma(k_g, r_g)$$

$$C_i \perp (B_i, T_{u,i}) \mid a_i, \quad Y_i = \min(T_i, C_i), \quad \delta_i = I(T_i \leq C_i)$$

$$S_g(t) = \pi_g + (1 - \pi_g)\{1 - F_\Gamma(t; k_g, r_g)\}$$

## Which assumptions hold automatically

- A1–A3 hold by construction from the simulation mechanism above.
- A4 holds in the regular gamma-mixture regime if

$$\varepsilon < \pi_g < 1 - \varepsilon, \quad 0 < k_L \leq k_g \leq k_U, \quad 0 < r_L \leq r_g \leq r_U.$$

# Appendix: Gamma Mixture Cure Remaining Checks

## What still has to be checked

- A5 holds when follow-up is mature enough that cure fraction and uncured-survival distribution are identifiable; this is not automatic in intentionally short-follow-up stress tests.
- A6 holds if gamma is the unique KL minimizer in  $\mathcal{D}$ , so  $\Pr(D_n = \text{Gamma} \mid \mathcal{S}_n) \rightarrow 1$ .

## Boundary caveat

If  $\pi_g = 0$  exactly, then the cure-fraction parameter is on the boundary, so A4 fails for the regular cure-model score theorem.

$$\pi_g = 0 \implies \eta_0^{(d)} \notin \text{int}(\Theta_d).$$

In that no-cure case, Method 1 can still satisfy  $\Pr(\mathcal{B}_n) \rightarrow 1$ , but the regular cure-score limit needs a separate boundary argument.